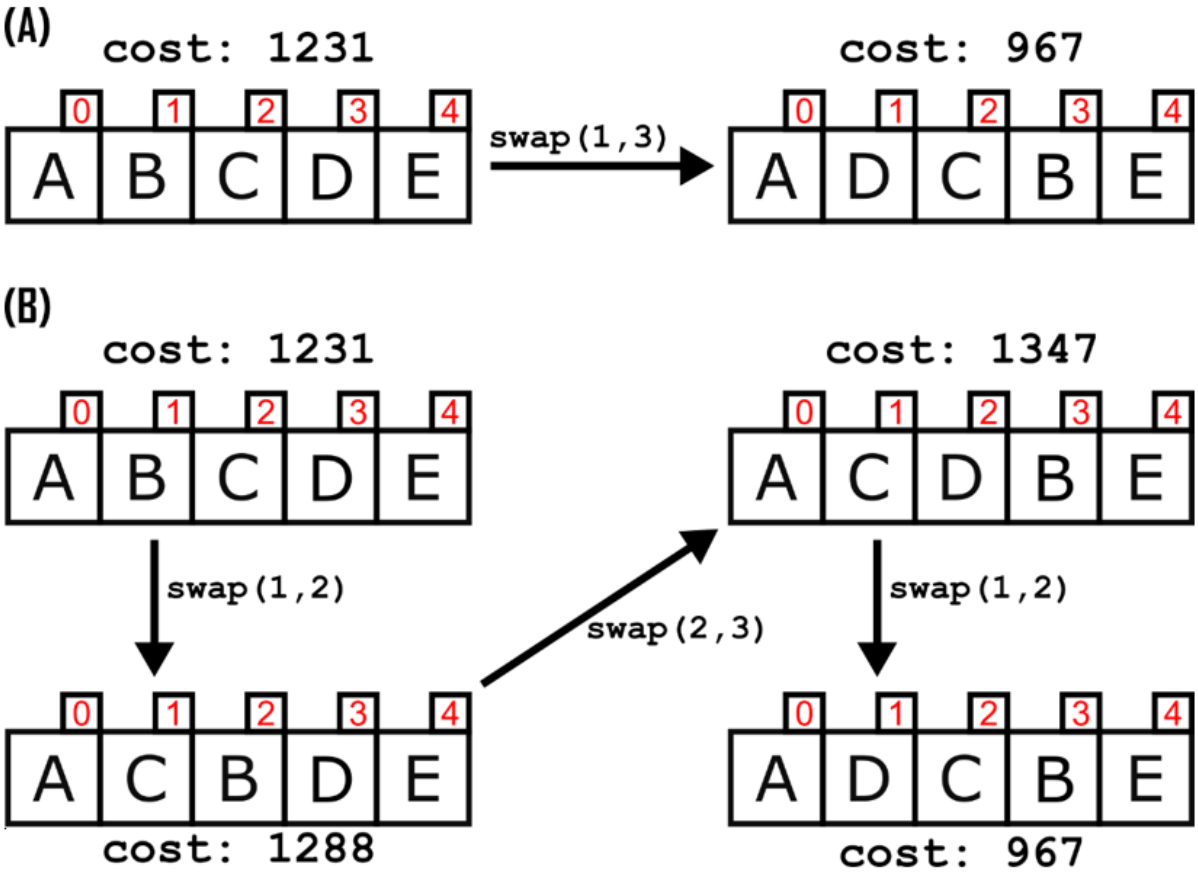


Figure 17.13 An example of why the local search transition heuristic, swapping only adjacent pairs, might struggle. In the top half, a swap between two random vertices (A) is performed: there is a good improvement in the cost function, so this transition will be always accepted.
When only swaps between adjacent pairs are allowed (B), several pejorative moves (2, in this case, but it can be made an arbitrarily long sequence) must be accepted before getting to the same configuration as (A). This might happen in the early stages of the cooling process, but as we get to the end of the cooling cycle it becomes unlikely.